



Lecture notes for Applies statistics, Probability theory

Lecture 8.

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Presentation Overview

Hypothesis
Testing

Parametric tests

Hypothesis Testing

Parametric tests

z-test

t-test



Section Overview

Hypothesis
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Hypothesis Testing



Steps of hypothesis test in practice

Hypothesis Testing

Parametric tests

- ① Learning the data
- ② Define a hypothesis (the hypothesis should be testable using the data)
- ③ Selecting a proper statistical test
- ④ Derive the distribution of the test statistic under the null hypothesis from the assumptions
- ⑤ Select a significance level, the maximum acceptable false positive rate
- ⑥ Compute from the observations the observed value
- ⑦ Decide to either reject the null hypothesis in favor of the alternative or not reject it



The hypothesis

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Tailedness of the alternative hypothesis test:

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Tailedness of the alternative hypothesis test:

- $H_1 : \theta < \theta_0$ or $H_1 : \theta > \theta_0$ is called one-tailed
- $H_1 : \theta \neq \theta_0$ is called two-tailed



Null and alternative hypothesis

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Definition

The

$$H_0 : \theta = \theta_0$$

is the null hypothesis and

$$H_1 : \theta > \theta_0 \text{ (or } H_1 : \theta < \theta_0 \text{)}$$

is the one-tailed alternative hypothesis.

If

$$H_0 : \theta = \theta_0$$

and

$$H_1 : \theta \neq \theta_0$$

then we talk about a two-sided alternative hypothesis.



Parametric tests

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Parametric tests

Hypothesis Testing

Parametric tests

Parametric tests are statistical tests that make assumptions about the parameters of the population distribution from which the sample is drawn.

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Some parametric tests:

- ① z-test
- ② t-test
- ③ F-test



Types of error

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Table of error types		Null hypothesis	
		True	False
Decision about null hypothesis (H_0)	Not rejected	True negative	Type II error
	Rejected	Type I error	False negative

Type I error - false positive

Type II error - false negative



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Concept question

We want to check if the weight of a bar of chocolate is equal to 100 grams. We weight $n = 16$ bars. The results are

$$x_1 = 100.3, x_2 = 99.8, \dots, x_{16} = 99.9$$

So it is our sample realization.

Notation. Let m denote the theoretical value of the weight. It is the expectation of the generic random variable X , i.e. $m = \mathbb{E}X$. X is the generic random variable of the underlying population.

$$X_1, X_2, \dots, X_n$$

is the sample from this population.



z-test

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Calculate the empirical mean. We know, that

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z-test

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So we standardize it, then we obtain a standard normal random variable

$$\frac{\bar{X} - m}{\frac{\sigma}{\sqrt{n}}} \sim \mathcal{N}(0, 1).$$



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$$z = \frac{\bar{X} - m}{\frac{\sigma}{\sqrt{n}}}$$



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$$z = \frac{\bar{X} - m}{\frac{\sigma}{\sqrt{n}}}$$

We emphasize that here we use m_0 . So z is standard normal if and only if $m = m_0$, i.e. the the null hypothesis H_0 is true.



Concept question solution

We denote the prescribed value of m by m_0 . In our example $m_0 = 100$.

$$H_0 : m = m_0$$

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We should decide if H_0 or H_1 is true.

In the case of z-test it is assumed that the sample comes from a normally distributed population with known variance. That is $X \sim \mathcal{N}(m, \sigma^2)$, where m is unknown, but σ is known.



The PDF of the t-distribution

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If $n \rightarrow \infty$, then the t-distribution with degree of freedom n converges to the standard normal distribution. Degrees of freedom, represented by df, is the number of independent information used to calculate a statistic. It's calculated as the sample size minus the number of restrictions.

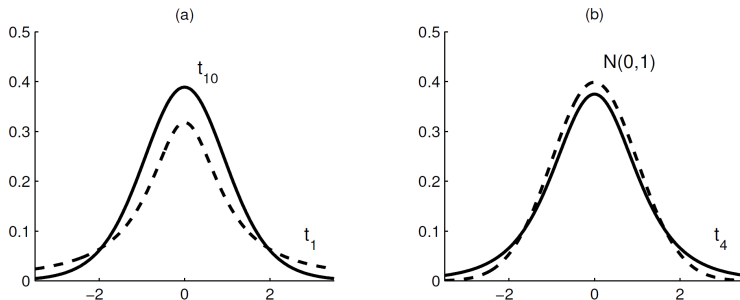


Fig. 1: PDF of t_1 and t_{10} (left) and the PDF's of t_4 and the standard normal PDF (right).



Concept question

Experiments with a new variety of tomato are being conducted at an agricultural research station. The crop is grown under carefully controlled conditions on ten experimental plots and the yields, in kg per plot, are found to be as follows:

60.2, 63.4, 58.8, 63.6, 64.7, 62.5, 66.0, 59.1, 65.1, 62.0;

It is known that an established variety of tomato would have a mean yield of 60 kg per plot on the experimental plot. Test, at the 95% level of significance, whether the new variety has a different mean yield, stating clearly your null and alternative hypotheses. State also the assumptions underlying your analysis.



Student's t-test

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The t-test is similar to the z-test. The major difference of the two tests is that in the case of t-test the variance is unknown. So let

$$X_1, X_2, \dots, X_n$$

be a sample from population $\mathcal{N}(m, \sigma^2)$, where m and σ are unknown. The null hypothesis is

$$H_0 : m = m_0,$$

where m_0 is a fixed number.

We start with the two-sided alternative hypothesis

$$H_1 : m \neq m_0$$



Student's t-test

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Calculate the empirical mean and the corrected empirical variance from the sample. From our previous theorem we know, that

$$\frac{\overline{X} - m}{\frac{s_n^*}{\sqrt{n}}}$$

has t-distribution with degree of freedom $n - 1$. So the test statistic of t-test is

$$t = \frac{\overline{X} - m}{\frac{s_n^*}{\sqrt{n}}}.$$

We emphasize that here we use m_0 . So t has t_{n-1} distribution if and only if $m = m_0$, i.e. the the null hypothesis H_0 is true.



The two tailed t-test

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Using significance level α , we reject H_0 if $|t| \geq t_{\alpha/2}$, where the critical value $t_{\alpha/2}$ satisfies $F(t_{\alpha/2}) = 1 - \frac{\alpha}{2}$ and F is the CDF of t-distribution with degree of freedom $n - 1$.



Concept question - solution

Standard deviation is unknown, so no z-test this time.

$$H_0 : \mathbb{E}X = 60 = m_0; \quad H_1 : \mathbb{E}X \neq 60$$

$$\frac{\bar{x} - m_0}{s_n^*} \sqrt{n} \sim t_{n-1}$$

$$\bar{x} = 62,54; \quad s_n^{*2} = \frac{1}{n-1} \left(\sum x_i^2 - n\bar{x}^2 \right) \approx 6.29; \quad s_n^* \approx 2.51$$

$$t = \frac{62.54 - 60}{2.51} \sqrt{10} \approx 3.2$$

Since $\alpha = 0.05$ and the alternative hypothesis is two-tailed, we use $\alpha/2 = 0.025$.

We know, that $t_{1-\alpha/2} = 2.2622$, so we reject H_0 , since $|t| > t_{1-\alpha/2}$



Two-sample t-test

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The two-sample t-test is a method used to test whether the unknown population means of two groups are equal or not.

Unpaired

Let X and Y be independent, also normally distributed random variables, $\mathbb{E}X = m_1, \mathbb{D}X = \sigma_1, \mathbb{E}Y = m_2, \mathbb{D}Y = \sigma_2$. $m_1, m_2, \sigma_1, \sigma_2$ are unknown. The two samples for X and Y are: $X_1, \dots, X_{n_1}$ and $Y_1, \dots, Y_{n_2}$. $df = n_1 + n_2 - 2$. The hypothesis is the following:

$$H_0 : m_1 = m_2; \quad H_1 : m_1 \neq m_2$$

Paired

Suppose two samples are matched pairs. Data from different pairs are independent, however, within each pair i , X_{0i} and X_{1i} are correlated. Hence the data in the control group and in the treatment group are correlated